SHIVAM KUMAR

Mechanical Engineering, DTU · Drones · Embedded · Robot vision

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SUMMARY

Second-year mechanical engineer who builds complete robots: **CAD and fabrication in Fusion 360, ESP32 firmware, and Pixhawk / Jetson autonomy**. Built the base station and landing gear for DTU ALTAIR's ISRO IRoC-U drone (3 rounds cleared) and led an SIH 2026 IoT early-warning system. Seeking internships in UAVs, robotics and robot vision.

EDUCATION

Delhi Technological University — B.Tech, Mechanical Engineering — AUG 2025 – JUL 2029

Coursework: Mechanics of Solids, Fluid Mechanics, Thermal Engineering, Engineering Materials, Applied Mathematics

DBRA SOSE School, Dwarka — Class XII (DBSE) — MAR 2025

School topper, 91.2% · two-year Robotics & Automation subject

ACHIEVEMENTS

- **ISRO IRoC-U 2026** — cleared 3 national rounds
- **Smart India Hackathon 2026** — cleared DTU internal round
- **Zonal science exhibitions** — 2 wins (school)
- **Technoxian 2026** — qualified for nationals
- **Delhi Robotics League** — Best Robot Design, 4th overall
- **World Forum for Industrial Innovation 2025** — delegate

PROJECTS

ISRO IRoC-U 2026 — Autonomous Drone & Base Station [3 ROUNDS CLEARED] — 2026 – PRESENT

`Fusion 360 · ABS & carbon fibre · Pixhawk / ArduPilot · Jetson Orin Nano · VIO · AprilTag`

- Designed the **base station and F450 landing gear** in Fusion 360; fabricated in 3D-printed ABS and carbon fibre to ±1 mm.
- Engineered a **pogo-pin charging interface and electromagnetic landing alignment** for hands-free docking and charging.
- Brought up the **Jetson Orin Nano stereo-camera pipeline** for visual-inertial odometry and AprilTag precision landing; resolved GStreamer/OpenCV conflicts and co-wrote the software design report.

Nexora-MineGuard — Mine Subsidence Early Warning [SIH 2026 · TEAM LEAD] — AUG 2026 – PRESENT

`ESP32 · ESP-NOW · LoRa · MPU6050 · flex & tilt sensors · FastAPI · React`

- Led a **6-member team**; wrote the system blueprint (architecture, BOM, wiring maps, JSON data contracts, power budget) so hardware, backend and ML sub-teams could build in parallel.
- Built a **4-node ESP32 mesh** over ESP-NOW with IMU, ball-tilt, vibration and flex sensing, bridged to a LoRa gateway, in 3D-printed enclosures.
- Swapped a planned LSTM for **explainable anomaly scoring** with 3-tier alerts; cut dashboard latency from ~15 s to 2–4 s.

Technoxian 2026 — PID Line Follower & 4WD RC Car [NATIONALS] — APR – AUG 2026

`ESP32 · PID · TB6612FNG · BTS7960 · FlySky · Fusion 360 · 3D printing`

- **Line follower:** 7-sensor IR array and TB6612FNG on an ESP32; Bluetooth-tunable PID with dt-normalisation, gain scheduling and hairpin detection.
- **RC car:** designed a single-motor 4WD differential chassis; firmware with 20 kHz BTS7960 drive, FlySky/Bluetooth control, cubic expo, ramp-limited throttle and failsafe.

Smart Ring Air-Mouse — Wearable BLE Controller — OCT 2026

`ESP32 · MPU6050 · NimBLE · BLE HID · capacitive touch`

- Ring that controls a laptop or phone as a **standard Bluetooth mouse**; one capacitive pad replaces three buttons; tuned EMA smoothing and a deadzone to remove IMU jitter.

Also: AAS CanSat 2027 (Team MetalHawk) — PocketQube and solar-panel release mechanisms, PDR · PathFinder-Q — vision-guided delivery bot proposal, IIITB × Qualcomm COMET Hackathon · NXP Cup India 2026 — ROS 2 + Gazebo simulation stack · Delhi Robotics League task robot — chassis, ball shooter and grabber arm.

LEADERSHIP & OUTREACH

DTU ALTAIR — Mechanical & Electronics Member — OCT 2025 – PRESENT

- Handle mechanical design and electronics for the team's competition builds; ran the fresher recruitment campaign and interviews.

Robotics Workshop, RPVV Dwarka — Co-instructor — AUG 2026

- Taught school students AI basics, the sense–think–act loop and robotics careers; wrote the community engagement report.

SKILLS

DESIGN	Fusion 360 (advanced, Python API), AutoCAD, SolidWorks & ANSYS (learning), 3D printing, carbon fibre, soldering
EMBEDDED	ESP32, Arduino, Pi Pico; I2C/SPI sensors (IMU, flex, optical flow, LiDAR); BTS7960, TB6612FNG; LoRa, ESP-NOW, BLE
AUTONOMY	Pixhawk / ArduPilot, Jetson Orin Nano, VIO, AprilTag landing, PID control, ROS 2 & Gazebo (learning)
SOFTWARE	Python, C / C++, FastAPI, React, Git, Linux (Ubuntu), LaTeX